

Project (15%): Creating a web page from other web sites

Overview

The goal of this project is to gather the live information from the Internet to create new web contents based on the theme you choose. That's how web sites like *Google News* (<http://news.google.com>) work. You are requested to submit the codes and the output results.

1. Create a new class called Web. Put the new methods in this class.
2. Create a web page of the information you think important from at least **three other web sites** (3%).
3. Add **pictures, a link to the sound** file which will play the sound when the user clicks on the link, add something that will make **random, relevant comments** about your theme depending on the live contents (3%).
4. **Zoom in** on the image when you click or move the mouse over it. (1%).
5. Develop a program that prompts the user to enter a word for **spell checking**. The program should integrate an online dictionary through the website backend to verify the spelling and display the result on the webpage. You may use additional web development technologies to create dynamic and interactive web pages (2%).
6. Implement the above functionality through the **automatic generation of code using a large language model**. In addition, a comparative **analysis** is presented to evaluate the advantages and disadvantages of building a website from scratch versus generating it with a large language model (6%).

Required prompt:

When generating the web crawler code, the crawler must comply with responsible web scraping practices:

- Respect the website's **robots.txt** and terms of service.
- Do not bypass CAPTCHA, authentication systems, or anti-bot protections.
- Avoid excessive requests by implementing rate limiting and delays.
- Only access publicly available data.
- Do not attempt to hack, exploit, or crack any website.
- The generated code should prioritize compliance, transparency, and server-friendly behavior.